Heart Rate Variability Analysis for LabVIEW

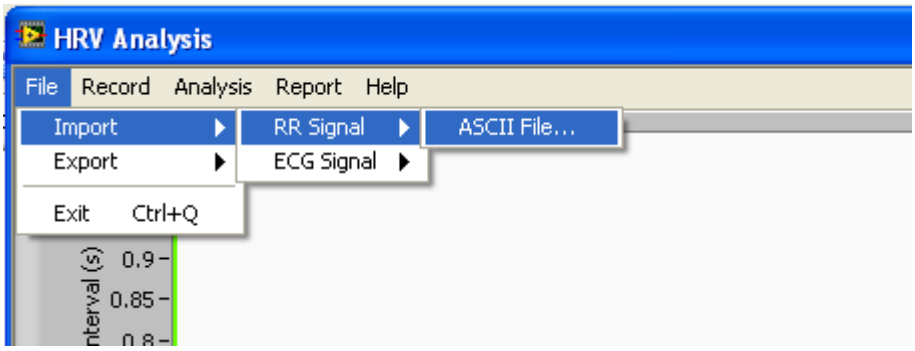
Quick Start

1 Introduction

This is a tool for Heart Rate Variability (HRV) Analysis. Heart Rate Variability which is defined by the variation between sequent heart beat rate or heart beat interval is a sign of heart health and autonomic nervous system.

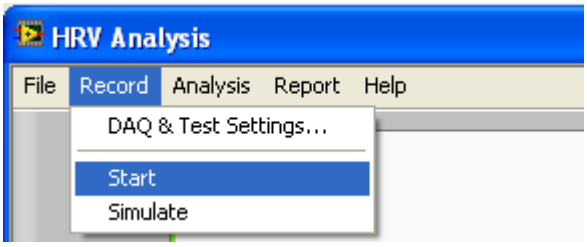
2 Data Source

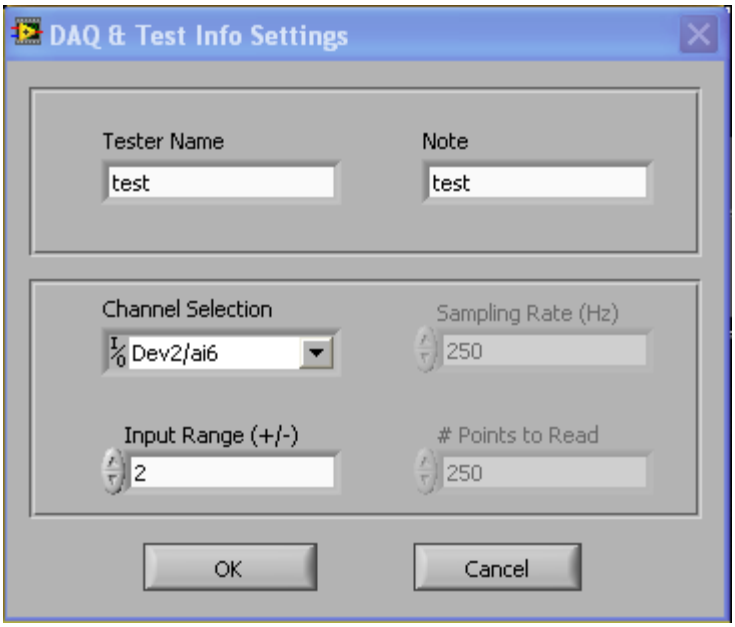
A. You can just import the existing heart beat intervals (RR intervals) signal to the program.

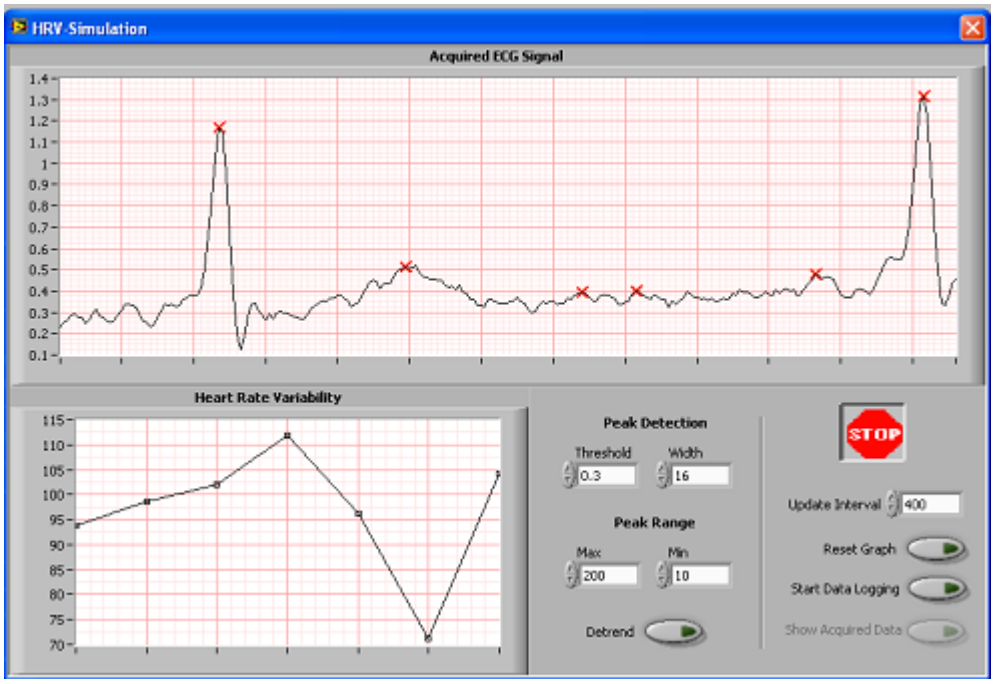


The file type is ASCII Text and only one column of RR interval values.

B. You can acquire ECG signals from ECG recorders using NI Hardware Device, and then extract the RR intervals.

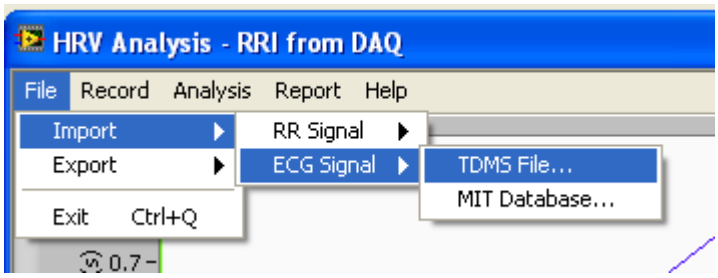




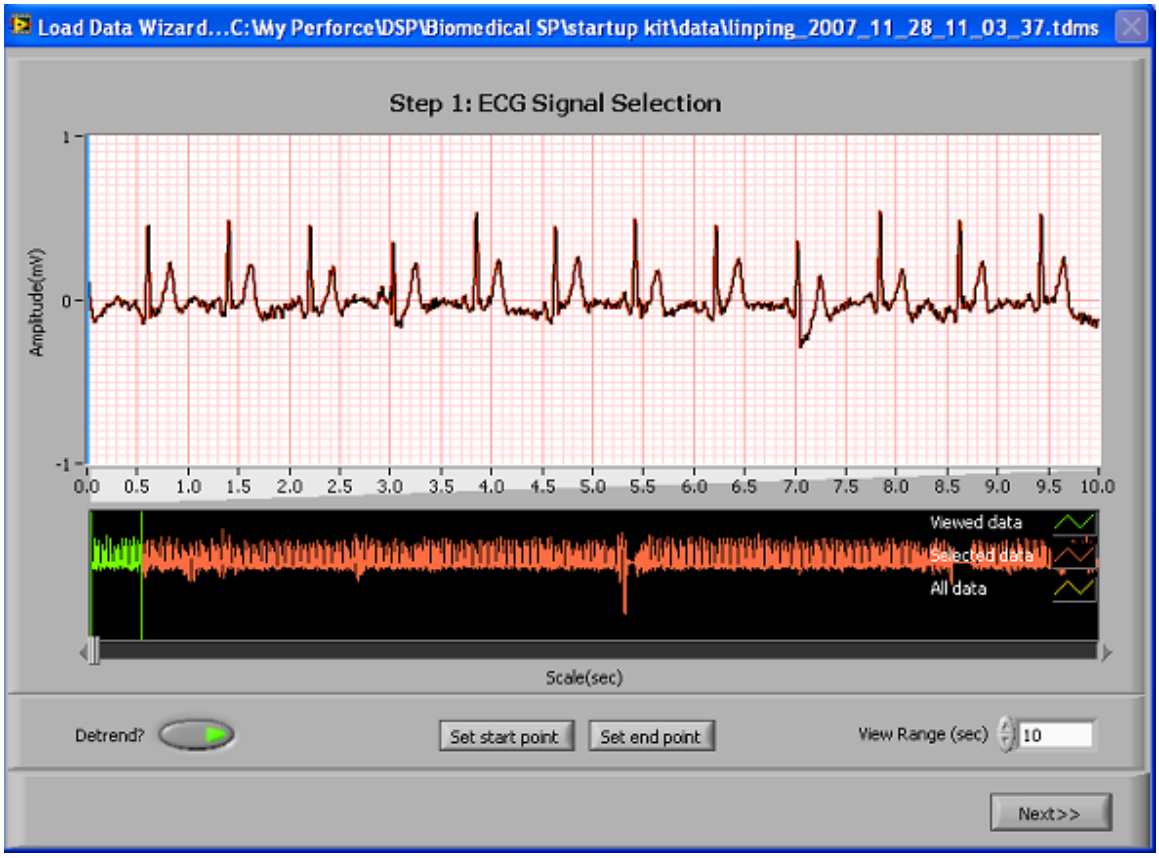


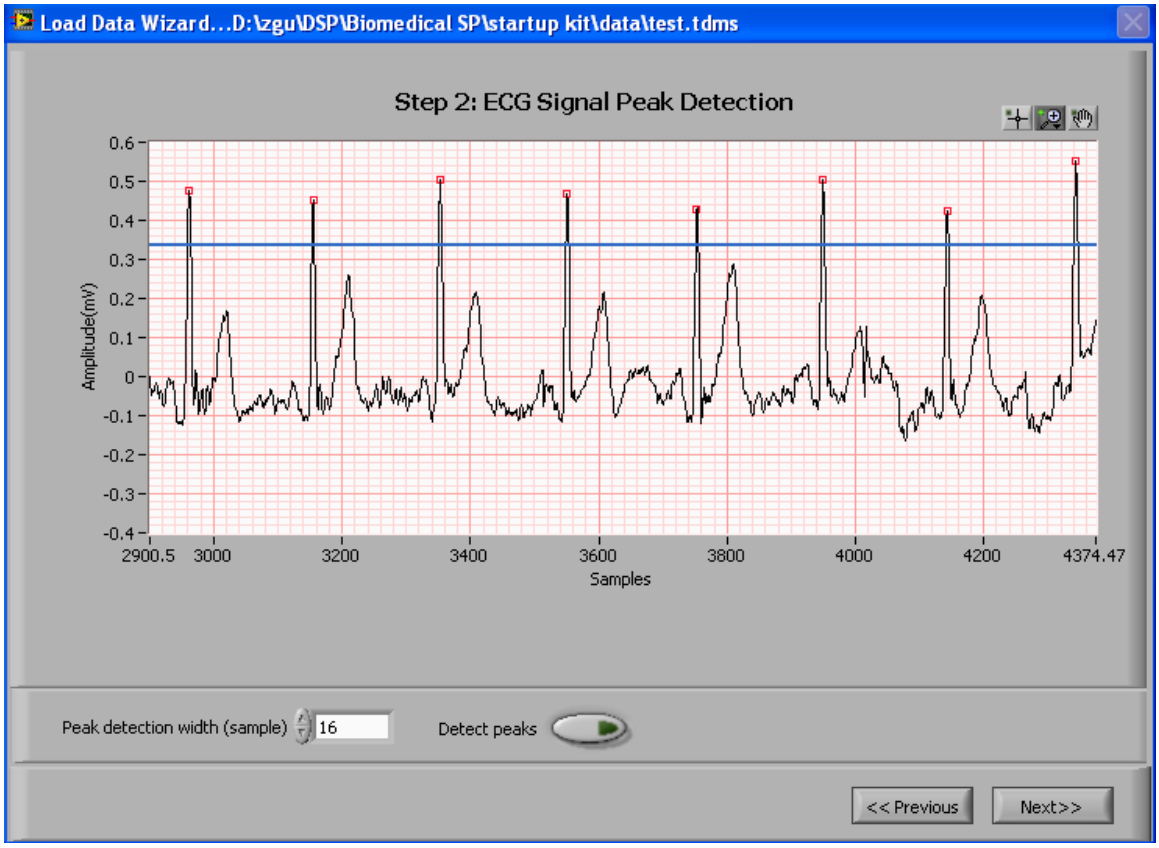
In the monitor window, you can inspect the online heart rate. The program will calculate the RR intervals automatically and then update the analysis result in the main window. You can also click "Start Data Logging" for writing the signal into a TDMS file.

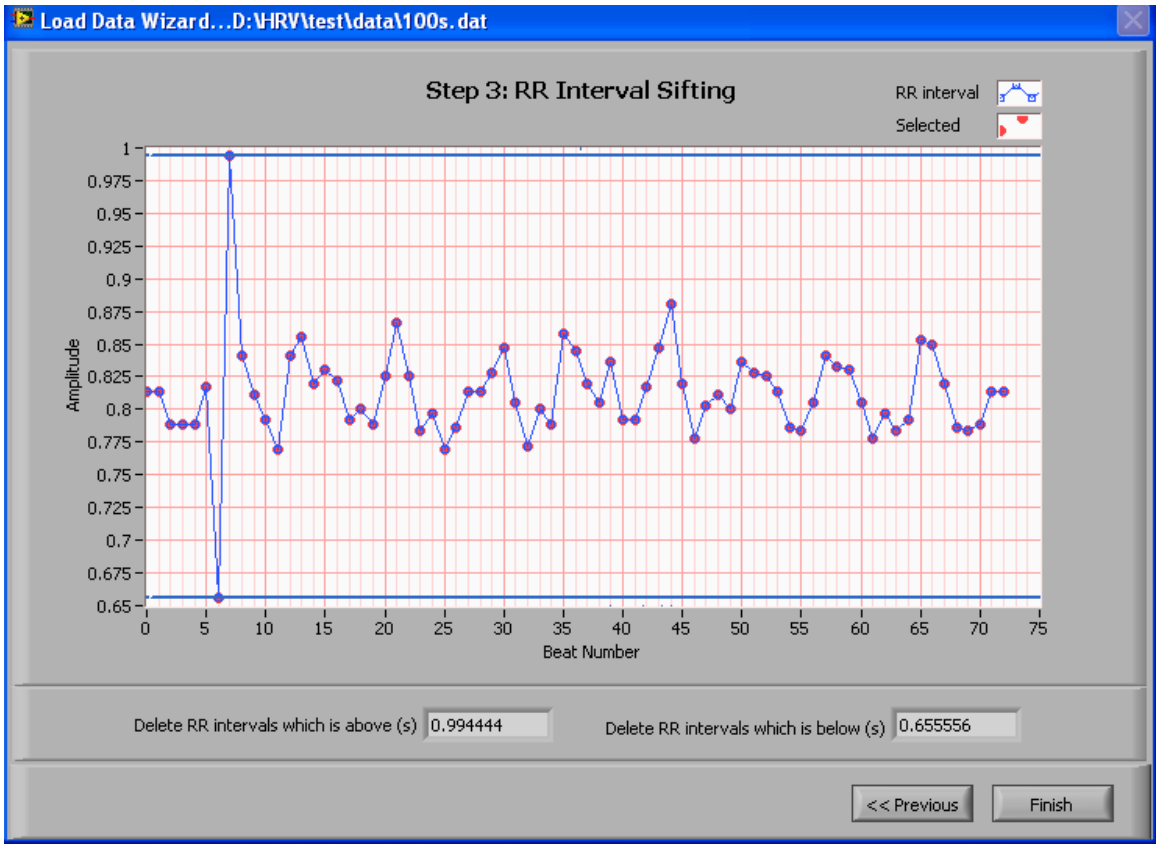
C. You can also import the data from MIT-BIH Database or the recorded TDMS file.



There is a wizard opened for you to import those files and guide you to extract RR signal step by step.

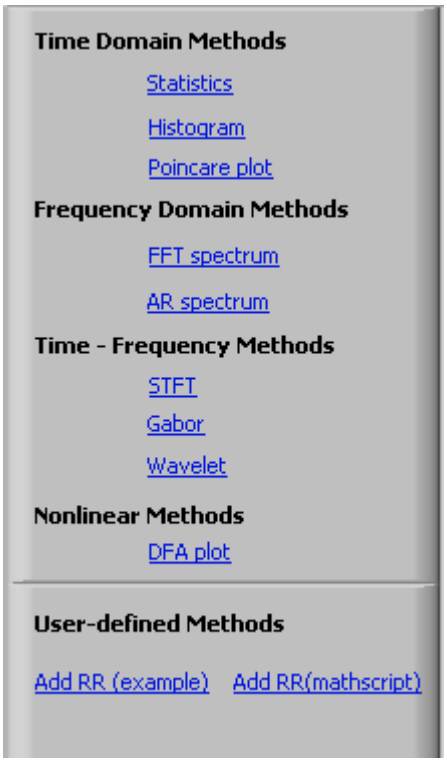



Load Data Wizard...D:\zgu\DSP\Biomedical SP\startup kit\data\test.tdms
Step 2: ECG Signal Peak Detection
Amplitude(mV)
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
2900.5
3000
3200
3400
3600
3800
4000
4200
4374.47
Samples
Peak detection width (sample)
16
Detect peaks
<< Previous
Next>>


Load Data Wizard...D:\HRV\test\data\100s.dat
Step 3: RR Interval Sifting
RR interval
Selected
Amplitude
1
0.975
0.95
0.925
0.9
0.875
0.85
0.825
0.8
0.775
0.75
0.725
0.7
0.675
0.65
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
Beat Number
Delete RR intervals which is above (s)
0.994444
Delete RR intervals which is below (s)
0.655556
<< Previous
Finish

3 Analyses

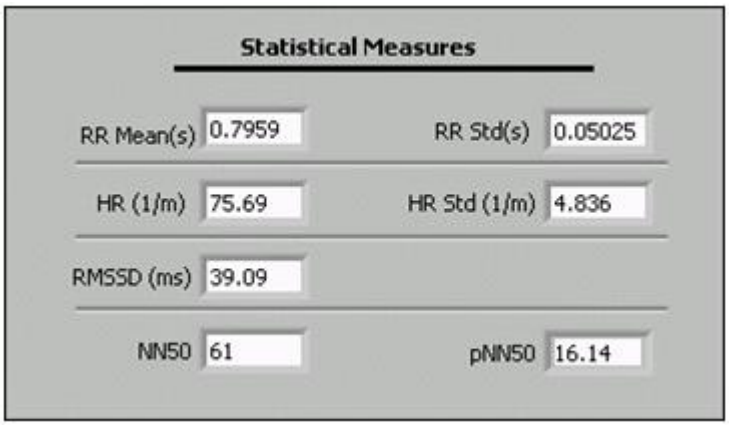
The application offers many methods to analyze the RR intervals such as time-domain analysis, spectral analysis and so on. You can just click the name of the method to see the analysis result by that mean.

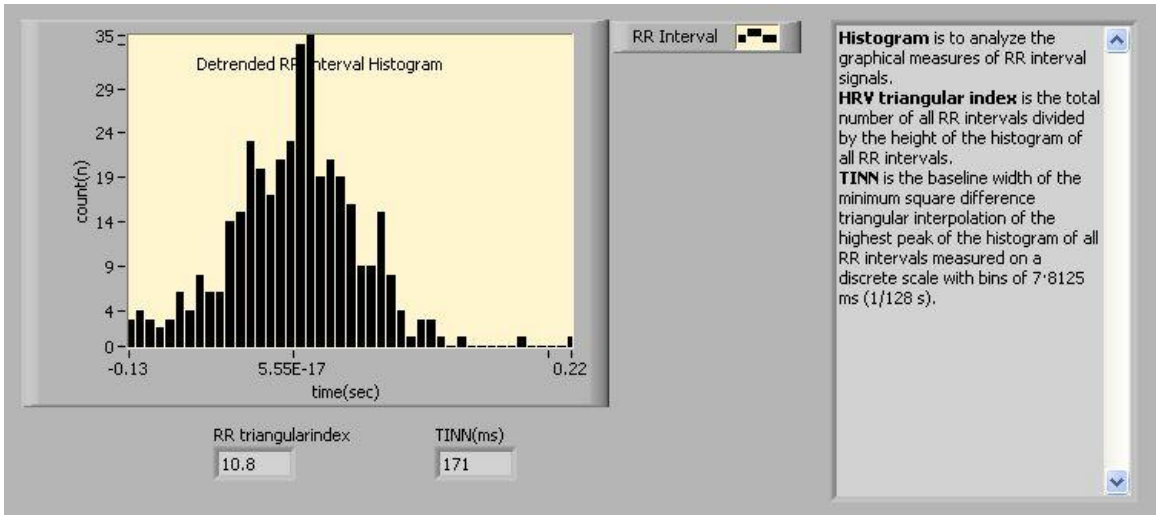


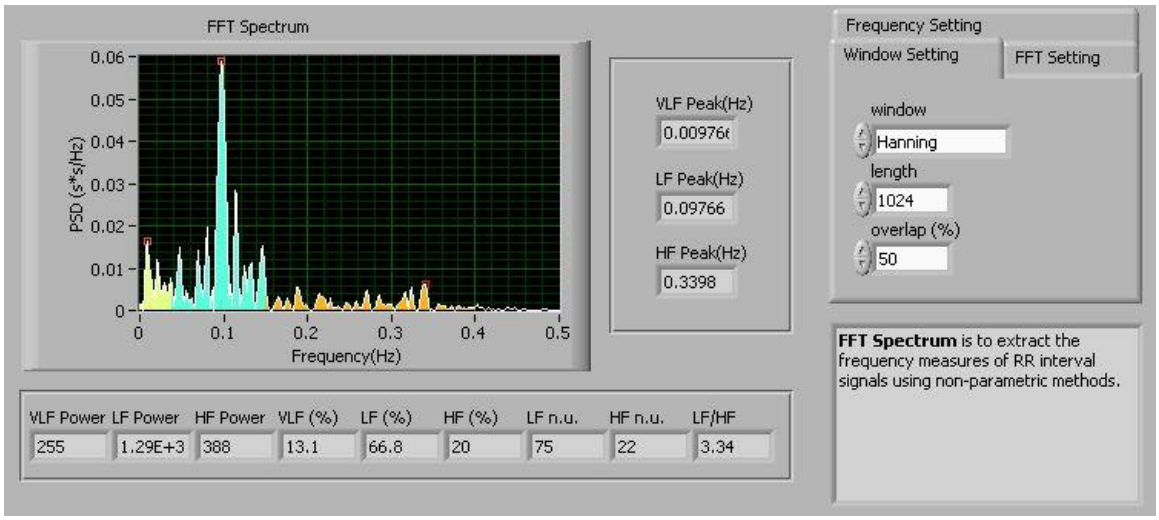
Online Analysis

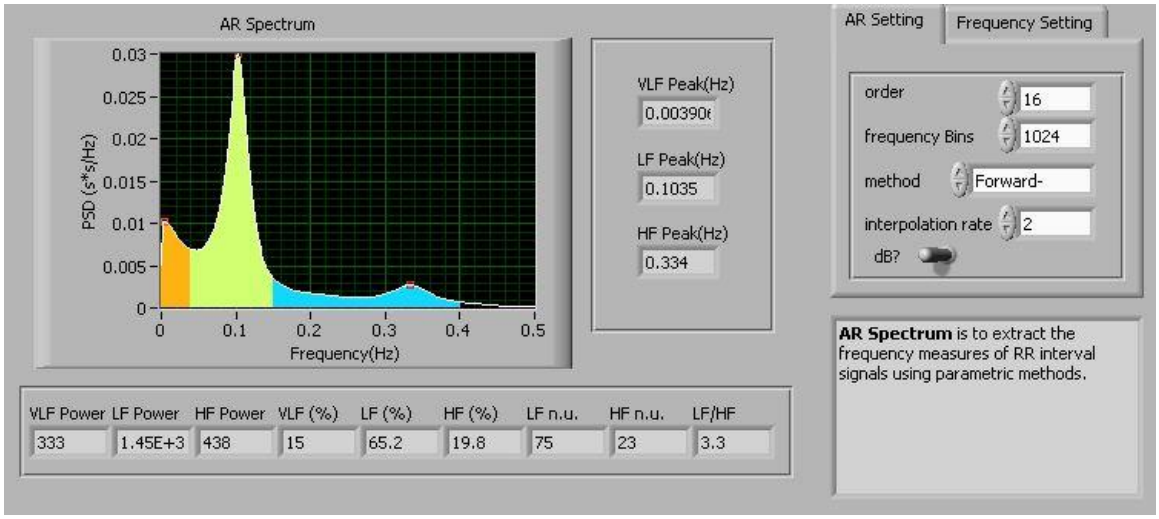
When acquiring ECG signals using NI Hardware Devices, the R peak of the ECG signals will be detected automatically by the peak detection settings. The RR intervals in the main window will change simultaneously, thus the analysis result will be updated.

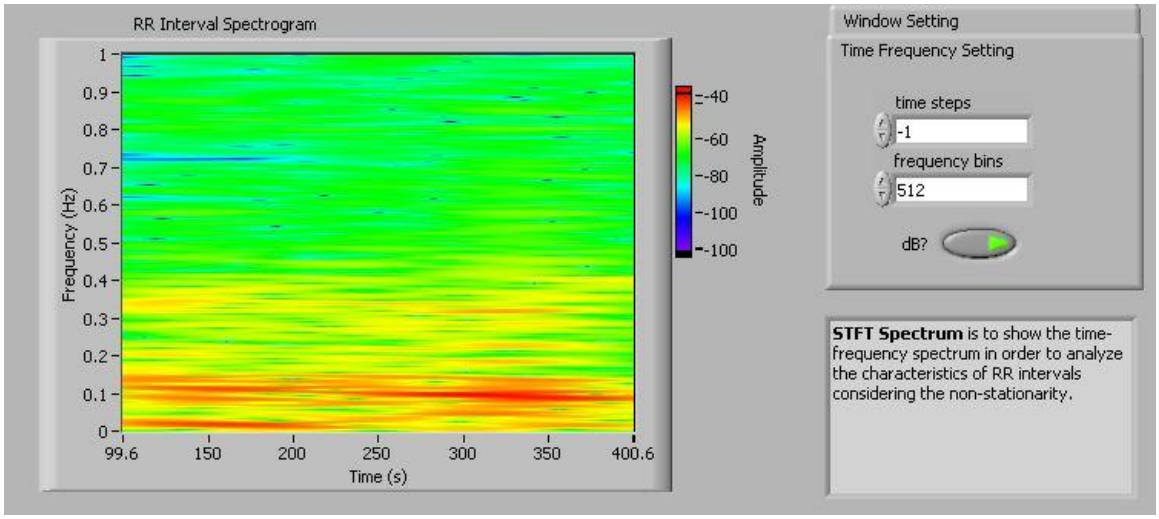
Here are some example analysis results.

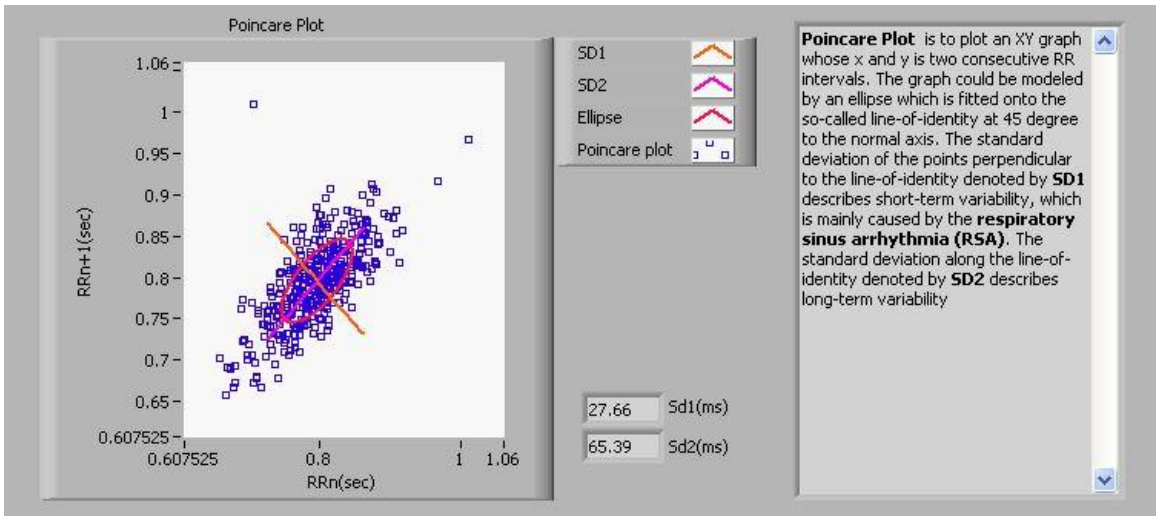


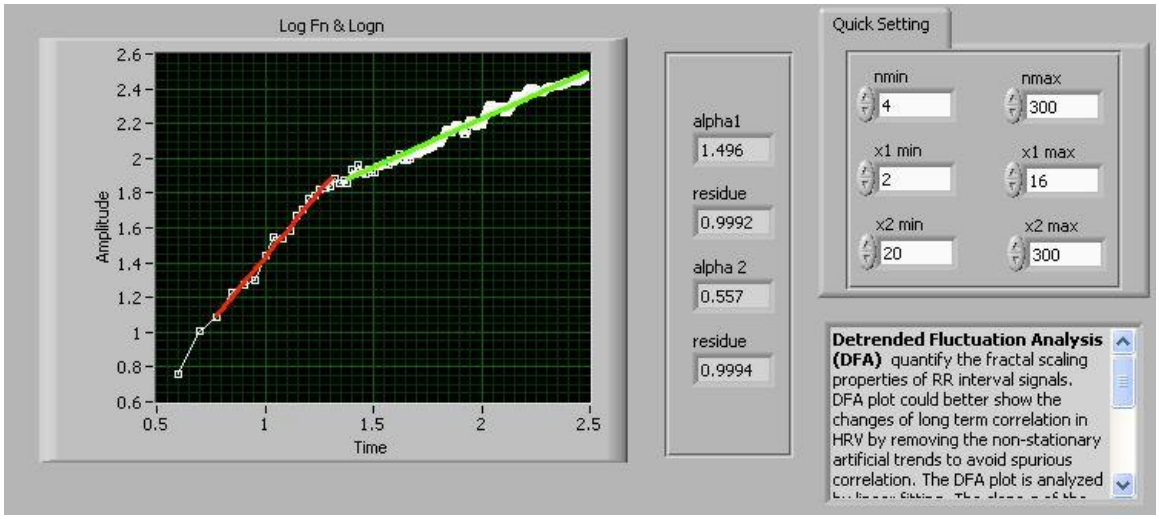

Detrended RR Interval Histogram
count(n)
time(sec)
RR Interval
RR triangularindex
10.8
TINN(ms)
171
Histogram is to analyze the graphical measures of RR interval signals.
HRV triangular index is the total number of all RR intervals divided by the height of the histogram of all RR intervals.
TINN is the baseline width of the minimum square difference triangular interpolation of the highest peak of the histogram of all RR intervals measured on a discrete scale with bins of 7·8125 ms (1/128 s).


FFT Spectrum
PSD (s*s/Hz)
Frequency(Hz)
VLF Peak(Hz)
LF Peak(Hz)
HF Peak(Hz)
Frequency Setting
Window Setting
FFT Setting
window
Hanning
length
1024
overlap (%)
50
VLF Power LF Power HF Power VLF (%) LF (%) HF (%) LF n.u. HF n.u. LF/HF
255 1.29E+3 388 13.1 66.8 20 75 22 3.34
FFT Spectrum is to extract the frequency measures of RR interval signals using non-parametric methods.

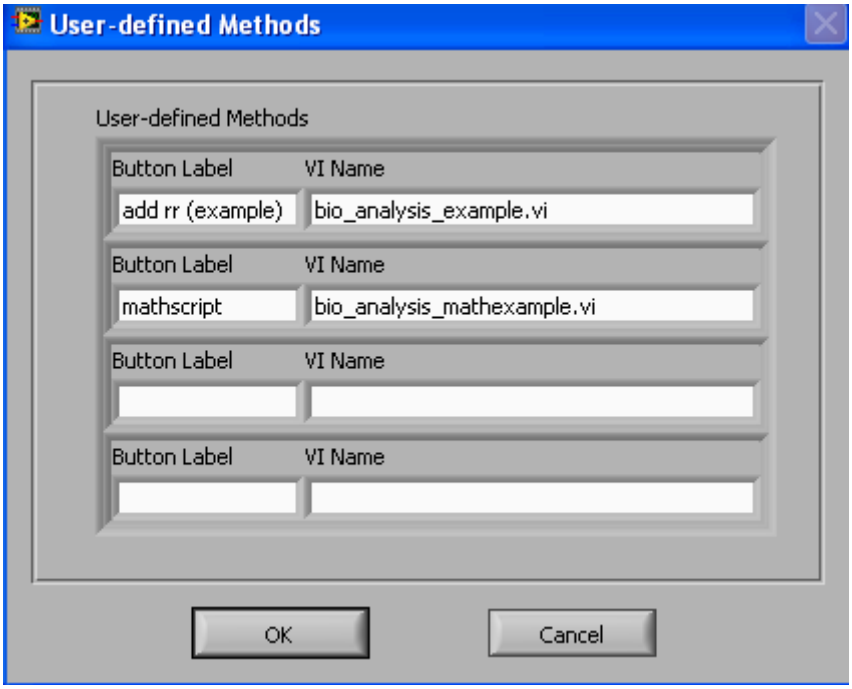

AR Spectrum
PSD (s*s/Hz)
Frequency(Hz)
VLF Peak(Hz)
LF Peak(Hz)
HF Peak(Hz)
AR Setting
Frequency Setting
order
16
frequency Bins
1024
method
Forward-
interpolation rate
2
dB?
VLF Power LF Power HF Power VLF (%) LF (%) HF (%) LF n.u. HF n.u. LF/HF
333 1.45E+3 438 15 65.2 19.8 75 23 3.3
AR Spectrum is to extract the frequency measures of RR interval signals using parametric methods.







You can also define your own analysis method using "User-defined Methods".

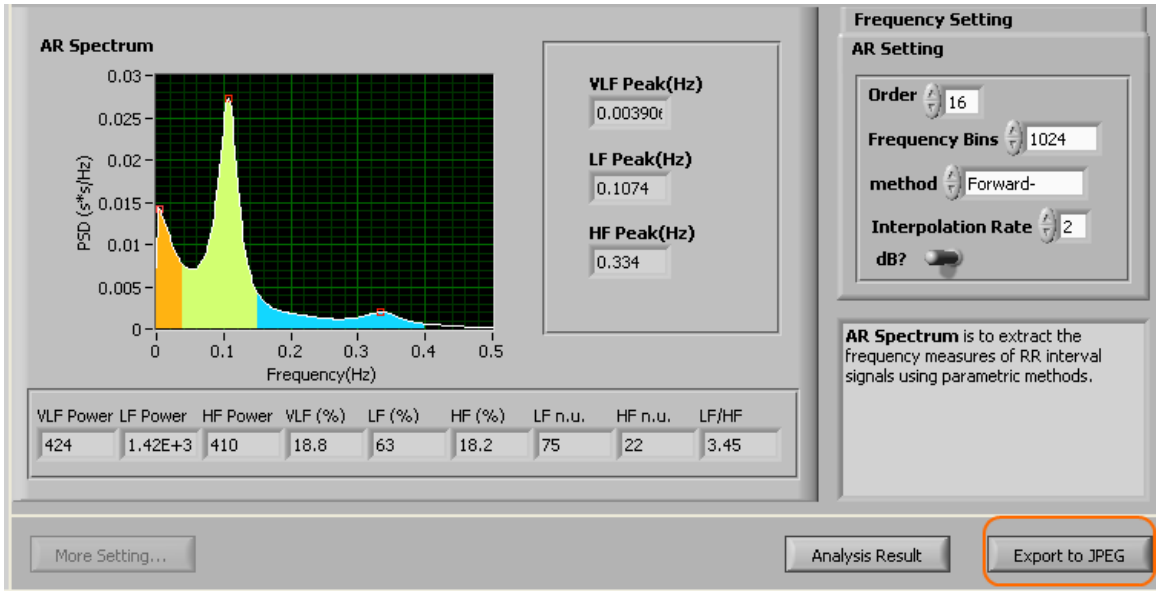


There are some VI templates under analysis folder. You can easily develop your own analysis method using LabVIEW G code or MathScript.

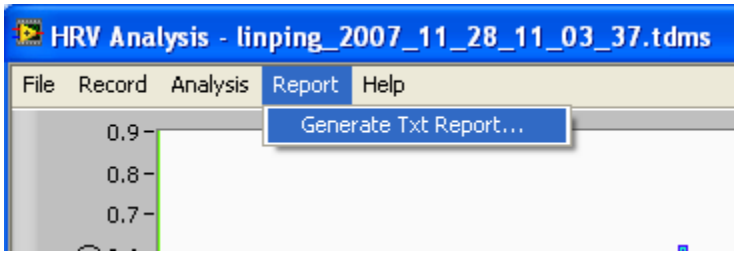
4 Results

Most analysis results have a plot and some feature measures. There are two ways to save the analysis results for future use.

A. Export the plot



B. Export the result as a Text File



HRV_linping_2007_11_28_11_03_37.tdms.txt - Notepad

File Edit Format View Help

```
Heart Rate Variability Analysis Report Text File
Made by Labview.
=======================
File Name:C:\My Perforce\DSP\Biomedical SP\startup
kit\data\linping_2007_11_28_11_03_37.tdms
Report Time:2/2/2008

Statistical Measures:

RR Mean(s): 0.27
RR Std(s) : 0.14
HR (1/m) : 300.48
HR Std (1/m): 332.86
RMSSD (ms): 226.99
NN50: 655.00
pNN50: 89.12

Histogram Measures:

RR Triangular Index: 0.00
TINN(ms): 73.31

Poincare Plot:

Sd1(ms): 160.62
Sd2(ms): 121.80

FFT Spectrum:

VLF Peak(Hz): 0.00
```

All the reported files are saved in folder export.